Name:

CHE 321; Spring – 2020; Quiz – 2   **100 Points**

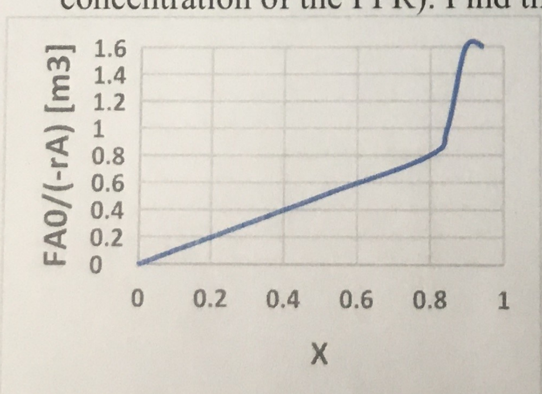
1 correct +8

1 wrong -3

1. (15 points) A first order reaction is carried out in 2 equal sized reactors. Select all options that are true about the overall conversion of the reaction:
   a. higher with 2 PFRs in series than with 2 PFRs in parallel. → wrong
   b. Changes with the order of PFR and CSTR (PFR-CSTR or CSTR-PFR).
   c. Remain the same for 2 PFRs in series or 2 PFRs in parallel.
   d. lower with two CSTRs in series then with two CSTRs in parallel. → wrong
2. (15 points) The most **unsuitable** reactor for carrying out reactions in which high reactant concentration favors high yields is
   a. series of CSTR
   b. one CSTR
   c. plug flow reactor
   d. series of PFRs
3. (15 points) 'N' plug flow reactors in series with a total volume 'V' gives the same conversion as a single plug flow reactor of volume 'V' for ____________ order reactions.
   a. third
   b. first
   c. any
   d. second
4. (15 points) The ratio of volume of CSTR to the volume of P.F.R. (for identical flow rate, feed composition and conversion) for zero order reaction is
   a. 1
   b. ∞
   c. 0
   d. >1
5. (15 points) Explain the answer above

   doesnt depend on concentration

6. (25 points) The following Levenspiel plot shows the response of a chemical reaction. A PFR and then a CSTR in series show a conversion of 0.4 and 0.8 (based on the entry concentration of the PFR). Find the volumes of the PFR and the CSTR.



0.8

0.3